

Junui Hong

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SUMMARY

SFU Software Systems graduate with a robust foundation in **C++** and **Backend Systems**. Proven track record in delivering real-world full-stack solutions using **Next.js** and **TypeScript**. Actively bridging the gap between development and operations by implementing **cloud-native architectures** (AWS) and **CI/CD** pipelines. Dedicated to engineering scalable, high-performance applications with production-grade reliability.

EDUCATION

Simon Fraser University | BSc in Computing Science – Software Systems, Dean's List (2 semesters) **May 2022 – Apr 2025**

SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, SQL, Java, Rust, Haskell

Frontend: Next.js, React, Tailwind CSS, HTML5/CSS3

Backend: Node.js, Express.js, MongoDB (Mongoose), Spring Boot, gRPC, RESTful API, OpenAI API, FastAPI

Tools & Systems: Linux (Ubuntu/Debian), Git, Docker, CMake, Postman, GoogleTest, OpenCV, AWS, Terraform

PROJECTS

Currency Insight Tracker V2 | FastAPI, AWS, Terraform, Docker | [GitHub](#) **April 2026 – May 2026**

- **Architectural Migration:** Engineered a transition from Serverless (Lambda) to a **containerized microservice** architecture to eliminate cold starts and ensure predictable, low-latency performance for backend APIs.
- **Infrastructure as Code (IaC):** Leveraged **Terraform** to define and deploy a modular cloud infrastructure, including **VPC, Private Subnets, and Application Load Balancer (ALB)**, ensuring 100% environment parity between development and production.
- **CI/CD & DevOps:** Orchestrated a fully automated pipeline using **GitHub Actions** to handle **Docker** multi-stage builds, image versioning in **Amazon ECR**, and seamless rolling updates to **AWS ECS**.
- **Security & Scalability:** Implemented a secure network topology by isolating application instances in private subnets and configuring **IAM roles** with the Principle of Least Privilege (PoLP) for enhanced system integrity.

Doum Academy Platform | Next.js, Express.js, MongoDB, TypeScript | [GitHub](#) | [Link](#) **Nov 2025 – Present**

- **Real-world Impact:** Engineering a full-stack platform to digitize curriculum management for a local academy, translating complex client requirements into optimized user interfaces and functional specifications
- **Self-Driven Stack Expansion:** Independently mastered **React and Next.js (App Router)** to bridge the gap between backend logic and modern frontend UX, achieving a fully responsive and optimized user interface
- **Architecture & API Design:** Developed a robust backend using **Express 5.2** and **Mongoose**, implementing secure **RESTful APIs** for dynamic content handling and automated email notifications via **Nodemailer**. Successfully managed evolving project scopes and client feedback cycles, ensuring code maintainability and scalability through strict **TypeScript** implementation

Social Gaming Platform | C++, WebSocket++, GoogleTest, CMake | [GitHub](#) **Sep 2024 – Dec 2024**

- **Agile Leadership & System Engineering:** Led a 7-person team as Scrum Master to build a high-concurrency multiplayer server using **WebSocket++**, ensuring real-time state synchronization and architectural consistency
- **Quality Assurance:** Integrated **GoogleTest** for automated unit testing, achieving comprehensive coverage of rule-based edge cases and significantly reducing post-deployment bugs

EXPERIENCE

STEM Tutor & Coding Instructor | The Doum Academy **Jan 2022 – Present**

- **Academic Excellence:** Delivered high-level instruction in advanced STEM subjects, including Calculus, Physics 11/12, and Pre-Calculus 11/12; maintained a 95%+ class average, demonstrating exceptional analytical and quantitative reasoning skills
- **CS Mentorship:** Provided technical guidance for **university-level coding courses** and Programming 11/12, mentoring students in fundamental data structures, algorithms, and logic building

Technical Assistant | DAEWON Info & Comm Corp. (South Korea) **May 2020 – Apr 2021**

- **System Troubleshooting:** Resolved 50+ weekly network and hardware issues for university workstations, ensuring zero downtime for academic operations through proactive maintenance

MILITARY EXPERIENCE / LEADERSHIP

Artillery Signal Operator | Republic of Korea Army (Sergeant) **Jun 2018 – Mar 2020**

- **Leadership & Problem Solving:** Commanded a 10-member squad, managing secure communication protocols and awarded **2nd place** in a brigade-level emergency restoration competition